

Trainings ▾

Preparatory Steps

The barring device shaft turns approximately two revolutions before the engine begins to turn. The device will **not** turn the engine opposite the direction of normal rotation.

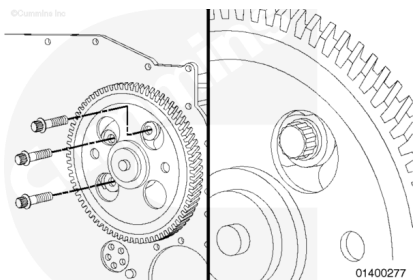
Push the shaft in and turn the barring device counterclockwise until the I-6TC mark on the pulley is aligned with the mark that is cast into the boss for the accessory drive seal on the front gear cover.



- Remove the gear cover and all related components. Refer to Procedure 001-031 in Section 1
(/qs3/pubsys2/xml/en/procedures/20/20-001-031-tr.html)
- Remove all of the cam follower assemblies. Refer to Procedure 004-001 in Section 4
(/qs3/pubsys2/xml/en/procedures/20/20-004-001-tr.html)
- Remove one injector using injector puller, Part Number 3823579, to allow timing to be checked. Refer to Procedure 006-026 in Section 6
(/qs3/pubsys2/xml/en/procedures/20/20-006-026-tr.html). Begin with number 3 cylinder
- Remove the camshaft idler gear. Refer to Procedure 001-036 in Section 1.
(/qs3/pubsys2/xml/en/procedures/20/20-001-036-tr.html)

Remove

Remove the three 12-point capscrews and thrust bearing plate.



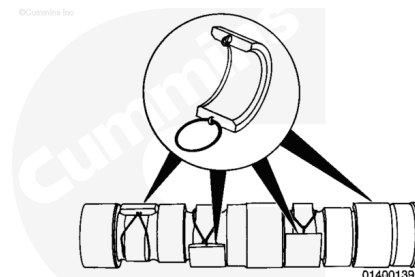
⚠ CAUTION ⚠

Pilots must be used to prevent damage to the camshaft and the bushings. Be sure the hooks do not damage the camshaft bushings.

For QSK19 CM500, use camshaft installation pilots, Part Number 3825150.

For QSK19 CM850 Modular Common Rail System (MCRS) and QSK19 CM2150 MCRS, use camshaft installation pilots, Part Number 5299377.

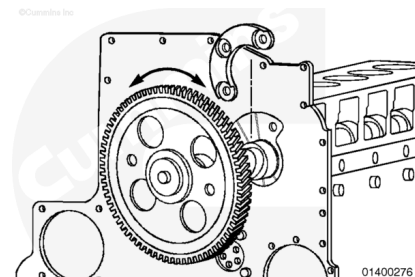
Place the pilots on the inner base circle of the valve lobes for the last two cylinders before removing the camshaft from the block.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lbs]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

Rotate the camshaft during removal so that one of the pilots is always on the bottom part of the camshaft bushing.



Clean

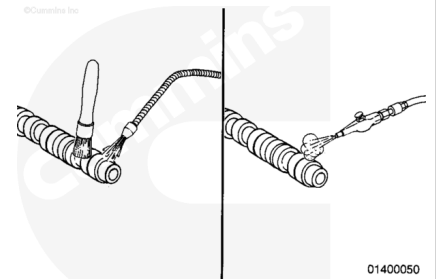
⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the

possibility of personal injury.

⚠ WARNING ⚠

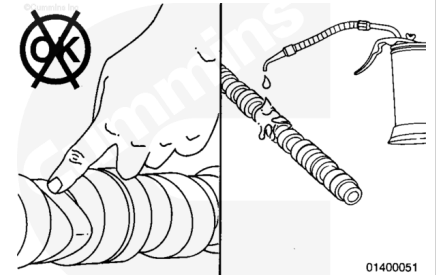
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Use solvent and compressed air to clean the camshaft.

After the camshaft has been cleaned, do **not** touch the machined surfaces with bare hands. This will cause rust to form.

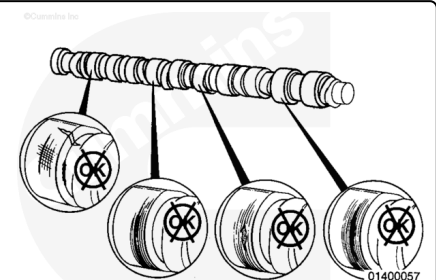
Lubricate the camshaft with clean 15W-40 oil before handling.



Inspect for Reuse

⚠ CAUTION ⚠

Do not repair the camshaft by grinding the valve or the injector lobes.



Inspect the camshaft.

Check the valve and the injector lobes for damage.

If the camshaft is damaged, it **must** be replaced.

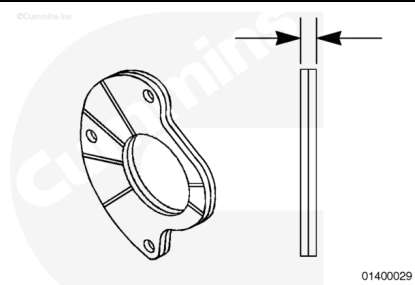
Measure each of the seven bushing journals.

Measure the outside diameter.

Camshaft Bushing Journal Outside Diameter		
mm		in
82.45	MIN	3.246
82.47	MAX	3.247

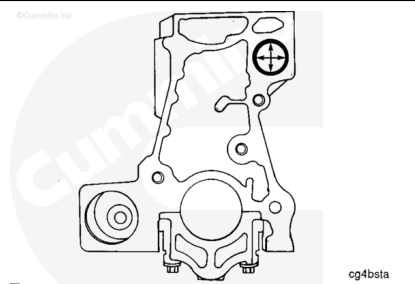
If the outside diameter is **not** within specifications, the camshaft **must** be replaced.

Inspect the camshaft thrust bearing. Refer to Procedure 001-056 in Section 1. (/qs3/pubsys2/xml/en/procedures/20/20-001-056.html)



Inspect the amount of copper exposure on the camshaft bushing. If it extends more than 180 degrees around the bushing, the bushing **must** be replaced.

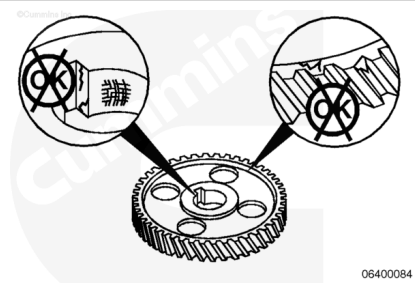
Measure the inside diameter.



Camshaft Bushing Inside Diameter		
mm		in
82.550	MIN	3.2500
82.600	MAX	3.2520

If the bushing is **not** within specifications, the bushing **must** be replaced.

If camshaft requires replacement, inspect the camshaft gear for reuse. Refer to Procedure 001-012 in Section 1. (/qs3/pubsys2/xml/en/procedures/20/20-001-012-tr.html)

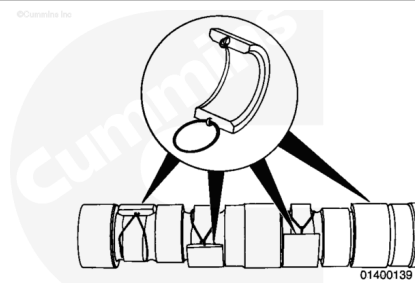


Install

For QSK19 CM500, install the camshaft installation pilots, Part Number 3825150, on the valve lobes of the camshaft.

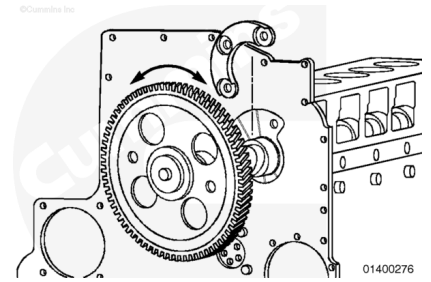
For QSK19 CM850 MCRS and QSK19 CM2150 MCRS, install the camshaft installation pilots, Part Number 5299377, on the valve lobes of the camshaft.

Use Lubriplate® 105, or equivalent, to lubricate the camshaft and camshaft bushings.



⚠ CAUTION ⚠

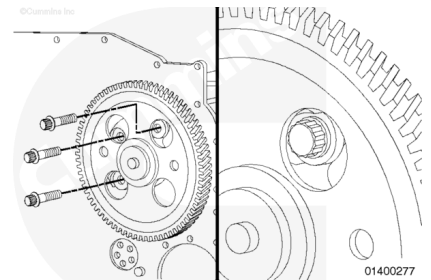
If the installation pilots are not available. Be careful when installing the camshaft to prevent the camshaft lobes from damaging the bushings.



Install the camshaft. Turn the camshaft backward and forward as it is being pushed. Keep the pilots turned downward to support the camshaft.

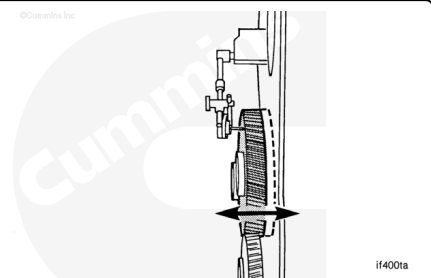
Tighten the three 12-point capscrews.

Torque Value: 64 n·m [47 ft·lb]



Use a dial indicator to measure the camshaft end clearance.

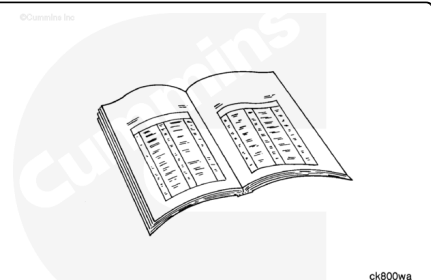
Camshaft End Clearance		
mm		in
0.15	MIN	0.006
0.33	MAX	0.013



If the clearance is **not** within specifications, check for foreign material or a piece of gasket between the thrust plate and the block.

Finishing Steps

- Install the camshaft idler gear. Refer to Procedure 001-036 in Section 1.
(/qs3/pubsys2/xml/en/procedures/20/20-001-036-tr.html) Make sure the index marks on the camshaft and crankshaft are in alignment with the timing marks on the idler gear.



Make sure the index marks on the camshaft and crankshaft are in alignment with the timing marks on the idler gear.

Measure the gear train backlash. Refer to Procedure 001-055 in Section 1 (</qs3/pubsys2/xml/en/procedures/20/20-001-055.html>)

The reason for installing **only** one cam follower assembly at this time is to save time if the camshaft has to be removed to adjust the injection timing.

Install the cam follower assembly on cylinder number 3. Refer to Procedure 004-001 in Section 4 (</qs3/pubsys2/xml/en/procedures/20/20-004-001-tr.html>)

Measure the injection timing. Refer to Procedure 006-025 in Section 6 (</qs3/pubsys2/xml/en/procedures/20/20-006-025-tr.html>)

Install the remaining cam follower assemblies and related components. Refer to Procedure 004-001 in Section 4 (</qs3/pubsys2/xml/en/procedures/20/20-004-001-tr.html>)

Install the gear cover and related components. Refer to Procedure 001-031 in Section 1 (</qs3/pubsys2/xml/en/procedures/20/20-001-031-tr.html>)